Dixit Patel [Ph.D.]

Senior Software Engineer - Artificial Intelligence (AI) | dbpatel8754@gmail.com

Website | Google Scholar

SUMMARY

Senior Software Engineer *with a strong focus on* ***applied AI systems and scalable software development****, combining rigorous academic research experience with hands-on industry delivery. Proven ability to design, build, and deploy production-grade AI-enabled applications using Python tech stack, cloud-based architecture, and modern machine learning and generative AI techniques. Experienced across the full software and AI lifecycle, including data preparation, model and LLM integration, backend system design, API development, testing, deployment, and performance optimization. Track record of translating complex research and engineering concepts into robust, real-world solutions such as AI-driven platforms, Retrieval-Augmented Generation (RAG) applications, and data-intensive backend services. Known for technical depth, system-level thinking, and ownership of complex features and architectures, with a collaborative approach to working with cross-functional teams and stakeholders. Consistently delivers reliable, scalable, and user-focused AI software solutions that create tangible business and operational impact.*

TECHNICAL SKILLS

Programming Languages:

Python, C#, Java, SQL, JavaScript, HTML, CSS

Artificial Intelligence (AI) & Machine Learning:

Generative AI (GenAI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Model Integration, Data Preparation and Evaluation, Python ML Libraries (such as NumPy, Pandas, Matplotlib), AI Agent Development, Prompt Engineering, AI System Evaluation and Validation, Inference Integration

Backend & Software Engineering:

RESTful API Development, Backend System Design, Microservices, Object-Oriented Design, Data Structures and Algorithms, System Design and Architecture

Databases & Data Technologies:

MySQL, PostgreSQL, MongoDB, JSON

Cloud & Deployment:

AWS, Docker, Scalable System Design, CI/CD Fundamentals

Frameworks & Platforms:

Django, .NET

Development Tools:

Git, GitHub, Microsoft Copilot Studio, Visual Code Studio, Unit Testing, Jupyter Notebook, Anaconda

Data & Pipelines:

Data Pipelines, ETL/Data Preprocessing, Data Quality & Validation

Operating Systems:

Linux, Windows, macOS

PROFESSIONAL EXPERIENCE

L&T Technology Services LLC [Client - Caterpillar Inc.]

Senior Software Engineer – Artificial Intelligence (AI), Mossville, IL, USA **May 2025 – Present**

- Design and develop AI-enabled software solutions to enhance engineering and business workflows, leveraging Python-based backend systems, cloud infrastructure, and modern generative AI techniques.
- Lead the design and implementation of Retrieval-Augmented Generation (RAG)-based web applications that enable secure, natural-language querying of enterprise customer data, significantly improving data accessibility and decision-making efficiency.
- Architect and integrate large language models into production applications, focusing on robust backend services, data pipelines, and access-controlled AI interactions.
- Build intelligent AI agents using Microsoft Copilot Studio to automate complex, multi-step tasks, reducing manual effort and streamlining client operations.
- Own testing, evaluation, and validation of AI models and AI-powered applications, ensuring accuracy, reliability, and adherence to defined performance benchmarks before production rollout.
- Collaborate closely with cross-functional teams-including software engineers, product managers, and stakeholders-to translate business requirements into scalable technical designs and delivered solutions.
- Contribute to continuous improvement of AI and software engineering practices by promoting code quality, maintainability, and deployment best practices across the development lifecycle.
- Stay current with advancements in generative AI (GenAI), large language models (LLMs), and applied AI systems, proactively identifying opportunities to enhance existing solutions and introduce new capabilities.

Fabricators, Inc.– Software Engineer, Chattanooga, TN, USA **April 2023 – May 2025**

- Designed and implemented Python-based backend solutions to support and optimize manufacturing operations, improving system efficiency, reliability, and maintainability.
- Engineered algorithmic and data-structure enhancements that improved execution performance by approximately 33%, demonstrating a strong focus on performance optimization and scalable software design.
- Developed and integrated RESTful backend services using Python, JSON, and relational databases (MySQL), enabling reliable data exchange across internal systems.
- Contributed to data-driven application workflows, including structured data processing, validation, and persistence, supporting operational analytics and decision-making.
- Conducted thorough peer code reviews and enforced coding standards to ensure high-quality, maintainable, and production-ready software.
- Managed source control and collaborative development using Git and GitHub, enabling efficient cross-functional teamwork and continuous integration.
- Applied strong software engineering fundamentals-object-oriented design, data structures, and algorithms-across the full development lifecycle to deliver robust and extensible solutions.

Wright State University – Graduate Research Assistant, Dayton, OH, USA **Dec 2018 – Dec 2022**

- Led design and development of applied software systems for mobile healthcare applications, leveraging Python and C# to support real-world healthcare workflows and improve care delivery.
- Translated research concepts into production-quality software solutions, delivering applications that exceeded client expectations and contributed directly to securing over $1M in grant funding for subsequent project phases.
- Designed and implemented comprehensive unit test suites to validate application functionality, ensuring robustness, reliability, and maintainability of deployed systems.
- Built backend components and integrated database systems using Python, C#, and relational databases (MySQL), supporting secure data storage, retrieval, and application logic.
- Applied strong software engineering principles including object-oriented design, modular architecture, and efficient data structures to develop scalable and extensible applications.
- Collaborated with multidisciplinary teams and external stakeholders, actively participating in technical discussions, project planning, and client communications to ensure alignment with project goals.
- Mentored and supported junior team members through technical guidance and knowledge sharing, contributing to a collaborative and productive research and development environment.

L&T Technology Services Limited – Software Engineer, Navi Mumbai, MH, India **June 2016 – June 2018**

- Contributed to the design and development of customized software solutions for enterprise clients using a Python-based technology stack, supporting end-to-end feature development and delivery.
- Played an active role in implementing and enhancing application features, working closely with senior engineers to ensure alignment with system requirements and performance expectations.
- Participated in the design and development of an AI model-based proof-of-concept (POC) for product quality control, applying Python libraries to support data analysis and analytical workflows.
- Developed, tested, and validated computer vision-based AI models to improve defect detection accuracy, contributing to enhanced product quality assessment.
- Supported software testing, validation, and documentation activities to ensure reliable, maintainable, and production-ready code.

EDUCATION

Doctor of Philosophy (Ph.D.) | Wright State University | 2019 - 2022

Computer Science and Engineering | GPA: 3.92/4.0

Master of Science (M.S.) | Wright State University | 2018 - 2022

Computer Science | GPA: 3.92/4.0

Bachelor of Engineering (B.E.) | Gujarat Technological University | 2011 - 2015

Electronics and Communication Engineering | GPA: 3.52/4.0

ACHIEVEMENTS

- **Outstanding Paper Award**, International Conference on Mobile Learning (IADIS), 2022, for research conducted during Ph.D. studies at Wright State University.

- **Finalist, Technical Paper Presentation Competition,** Technology Conclave – VI, organized by L&T Technology Services; topic: Digital Transformation.
- Awarded full financial support for both M.S. and Ph.D. degree programs at Wright State University in recognition of **research excellence and contributions** to applied healthcare technology projects.
- **Peer Reviewer Recognition** and digital badge for completion of 30+ peer reviews with JMIR Publications and Discover Artificial Intelligence, supporting advancements in engineering, research, AI, digital health, and open science.
- Awarded three **Certificates of Excellence** under the AICTE Employability Enhancement Training Programme (EETP) by Bharat Sanchar Nigam Limited (BSNL): [1] BSNL Silver Certified Engineer, [2] BSNL Gold Certified Engineer and [3] BSNL Platinum Certified Engineer.

RESEARCH PROFILES

1. ***Google Scholar Profile:*** *https://scholar.google.com/citations?user=Bw-PUssAAAAJ&hl=en*
2. ***ResearchGate Profile:*** *https://www.researchgate.net/profile/Dixit-Patel-18*

RECOMMENDATIONS

- Please refer to the Recommendations section on my *LinkedIn Profile.*